Design Your Playground!

Description:

Given the length and width in term of expressions, students will find the perimeter and area of different rectangles by distributive property and skills of simplifying expressions.

Standards:

7.EE-1

7.EE-3

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Fountain

x + 4 yd

10 yd

What is the perimeter of this block?

Add the four sides.

[ ] + [ ] + [ ] + [ ]

= Your simplified expression yd

What is the area of this block?

Multiply length and width.

[ ] × [ ]

= Your simplified expression yd^2

Check my answer

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Slide

3 yd

2x - 5 yd

What is the perimeter of this block?

Add the four sides.

[ ] + [ ] + [ ] + [ ]

= Your simplified expression yd

What is the area of this block?

Multiply length and width.

[ ] × [ ]

= Your simplified expression yd^2

Check my answer

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Swing

2.5 yd

x + 2 yd

What is the perimeter of this block?

Add the four sides.

[ ] + [ ] + [ ] + [ ]

= Your simplified expression yd

What is the area of this block?

Multiply length and width.

[ ] × [ ]

= Your simplified expression yd^2

Check my answer

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What is the perimeter of this block?

Add the four sides.

[ ] + [ ] + [ ] + [ ]

= Your simplified expression yd

What is the area of this block?

Multiply length and width.

[ ] × [ ]

= Your simplified expression yd^2

Check my answer

Statue

x - 3 yd

2 yd

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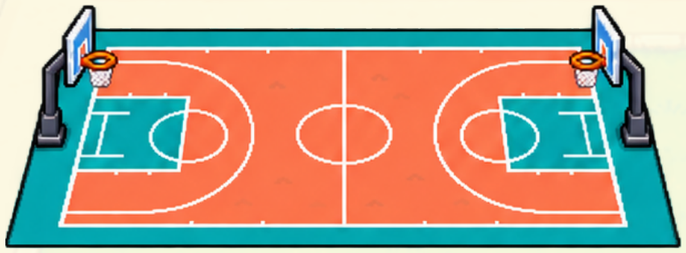
Basketball Court



2x yd

35 yd

What is the perimeter of this block?

Add the four sides.

[ ] + [ ] + [ ] + [ ]

= Your simplified expression yd

What is the area of this block?

Multiply length and width.

[ ] × [ ]

= Your simplified expression yd^2

Check my answer

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Mini Soccer Field

28.2 yd

4x + 6.4 yd

What is the perimeter of this block?

Add the four sides.

[ ] + [ ] + [ ] + [ ]

= Your simplified expression yd

What is the area of this block?

Multiply length and width.

[ ] × [ ]

= Your simplified expression yd^2

Check my answer

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Now name your own block!

Choose equipment for your playground. Drag or tap a picture below, then name your block here.

Name:

Moonbeam Carousel

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Moonbeam Carousel

What is the perimeter of this block?

Add the four sides.

[ ] + [ ] + [ ] + [ ]

= Your simplified expression yd

What is the area of this block?

Multiply length and width.

[ ] × [ ]

= Your simplified expression yd^2

Check my answer

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Drag or tap any 3 equipments into the playground.

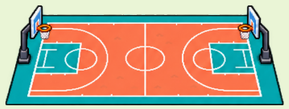
Then, place the picture of your own equipment in the playground too!

Fountain

Slide

Swing

Statue


Basketball Court

Mini Soccer Field

Moonbeam Carousel

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Let's finish up the plan! Find the area left for paths so the manager can plan the paving.

Area of this playground: (Show work)

Your simplified expression yd^2

Total area of your equipments: (Show work)

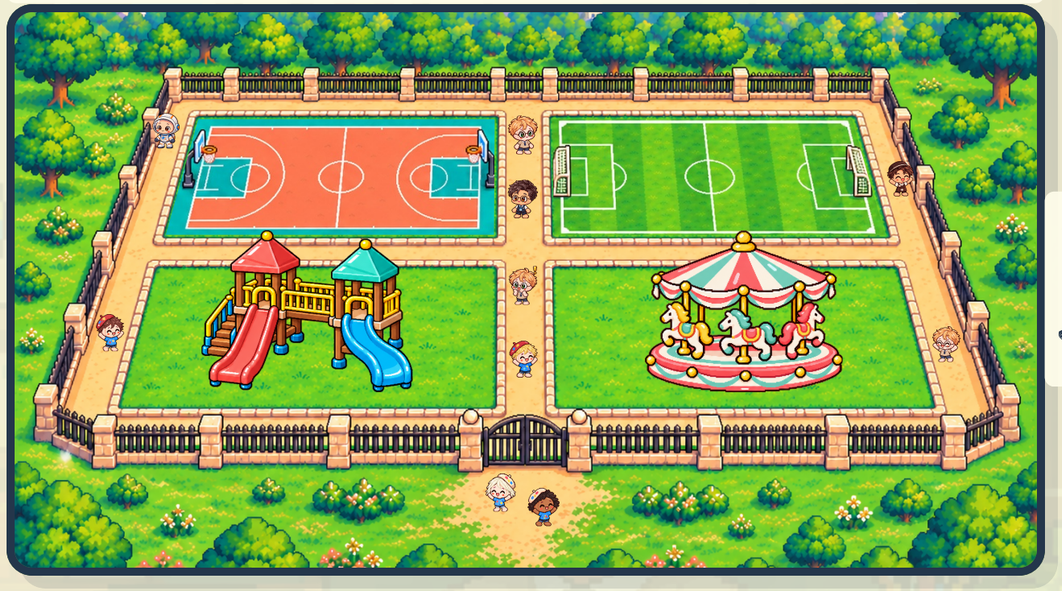
Your simplified expression yd^2

Remaining area for paths: (Show work)

Your simplified expression yd^2

Check my answer

$x + 32$ yd



400 yd

Equipment	Length (yd)	Width (yd)
Basketball Court	35	2x
Mini Soccer Field	4x + 6.4	28.2
Slide	2x - 5	3
Moonbeam Carousel	23	2x + 3y

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